Polylogarithmic Resolution Width for the Jordan-Curve Contradiction - medium

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1 Preliminaries

For an integer $q \geq 1$, write

$$[q] := \{1, \ldots, q\}, \qquad [q]_0 := \{0, \ldots, q-1\}.$$

Fix integers $d, m \geq 1$, and let

$$D := 2^d, \qquad M := 2^m, \qquad M \geq 2D.$$

For each timestamp $t \in [M]$, introduce two blocks R_t and B_t of $2d$ Boolean variables, interpreted as binary encodings of points in $[D]_0^2$. Thus the total number of variables is

$$N := 4Md.$$

For a $2d$-bit block $Z = (z_1, \ldots, z_{2d})$ and a point $p \in [D]_0^2$, let $\mathrm{bin}(p) \in \{0,1\}^{2d}$ be the binary encoding of p, and define the clause

$$\mathrm{NEQ}(Z, p) := \bigvee_{k\colon \mathrm{bin}(p)_k = 0} z_k \;\vee\; \bigvee_{k\colon \mathrm{bin}(p)_k = 1} \neg z_k.$$

This clause is falsified exactly when Z encodes p.

Definition 1 (Jordan-curve contradiction). *Let $\mathrm{JC}_{D,M}$ be the CNF consisting of the following clauses.*

1. *Unit clauses fixing the endpoints*

$$R_1 = (0,0), \qquad R_M = (D-1, D-1),$$

$$B_1 = (0, D-1), \qquad B_M = (D-1, 0).$$

2. *For every $C \in \{R, B\}$, every $t \in [M-1]$, and every $p, q \in [D]_0^2$ with $\|p - q\|_1 > 1$, the continuity clause*

$$\mathrm{NEQ}(C_t, p) \;\vee\; \mathrm{NEQ}(C_{t+1}, q).$$

3. *For every $i, j \in [M]$ and every $p \in [D]_0^2$, the disjointness clause*

$$\mathrm{NEQ}(R_i, p) \ \vee \ \mathrm{NEQ}(B_j, p).$$

Thus $\mathrm{JC}_{D,M}$ *asserts that two continuous grid curves joining opposite pairs of corners never meet. This CNF is unsatisfiable.*

Definition 2 (Resolution width)**.** *The Resolution rule is*

$$\frac{C \vee x \qquad E \vee \neg x}{C \vee E}.$$

A Resolution refutation of an unsatisfiable CNF F is a sequence of clauses ending with the empty clause, where every clause is either a clause of F or is obtained from earlier clauses by the Resolution rule. The width of a clause is its number of literals, and

$$\mathrm{ResWidth}(F) := \min_{\Pi} \max_{C \in \Pi} |C|,$$

where the minimum is over all Resolution refutations Π of F.

2 Problem formulation

Open Problem 1 (Polylogarithmic Resolution width)**.** *Does there exist a constant $c > 0$ such that, for all $d, m \geq 1$ with $M = 2^m \geq 2D = 2^{d+1}$,*

$$\mathrm{ResWidth}(\mathrm{JC}_{D,M}) \leq (\log N)^c, \qquad N = 4Md?$$